



# Psychometric properties of an Arabic translation of the Breast Appreciation Scale (BrAS) in Lebanese women

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## ABSTRACT

The 9-item Breast Appreciation Scale (BrAS) measures the construct of breast appreciation, posited as a facet of the broader positive body image construct. To date, there has been limited assessment of the psychometric properties of the BrAS across national and linguistic groups. To add to the literature on breast appreciation, we assessed the psychometric properties of an Arabic translation of the BrAS. An online convenience sample of 626 cisgender women (age range = 18–50 years) from Lebanon completed the Arabic BrAS along with additional measures of body image, disordered eating, and psychological well-being. Across two split-half subsamples ( $n = 312$  and  $n = 314$ , respectively), exploratory and confirmatory factor analysis showed that it was possible to extract a unidimensional model of the BrAS with all nine items. This model of the BrAS had adequate composite reliability in both split-half subsamples. Additionally, the BrAS evidenced good patterns of convergent (positive associations with other indices of positive body image and breast awareness, and negative associations with body dissatisfaction and symptoms of disordered eating) and concurrent validity (positive relationships with gratitude and compassionate self-responding). These results suggest that the Arabic translation of the BrAS retains its robust psychometric properties in Lebanese women.

## 1. Introduction

Although scholars generally agree that “breasted experiences” (cf. Young, 1992) play an important role in shaping (cisgender) women’s body image (e.g., Millsted & Frith, 2003; Swami & Furnham, 2018), relevant research has historically focused on breast size dissatisfaction (e.g., Lombardo et al., 2019; Swami et al., 2015; Swami et al., 2020) and broader body image outcomes in breast cancer survivors (e.g., Helms et al., 2008). While such research remains valuable, particularly in view of sociocultural discourse that continues to commodify, fetishise, and objectify the breasts (Gerald & Potvin, 2009; Swami & Tovée, 2013), the focus on negative experiences risks missing the complexities of breasted

experiences (Lee, 1997; Nash, 2014). For example, Young (1992) described how – from the perspective of women themselves and in the absence of a male gaze – women’s sensory experiences of their breasts (e.g., how they move, how sensitive they are) matter more than aesthetics. In addition, in women’s narratives, the breasts are also often identified as sites of pleasure (Young, 1992), sexual authority (Lee, 1997), and joy (McCann, 2020).

Motivated by the broader shift in the body image literature towards understanding aspects of positive body image (for reviews, see Daniels et al., 2018; Tylka & Piran, 2019), scholars have also recently begun to examine positive breasted experiences in cisgender women. Such experiences can be wide-ranging, encompassing feelings of maternal joy,

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child nurturance, and positive femininity derived from one's breasts (Millsted & Frith, 2003; Schmied & Barclay, 1999). However, one specific form of positive breasted experience that has received scholarly attention is *breast appreciation*, which Swami, Todd, and Tylka (2022, p. 277) defined as "accepting, holding favourable opinions toward, and respecting one's breasts and their functions, while also rejecting media-promoted breast ideals as the only form of beauty". In the view of Swami, Todd, and Tylka (2022), breast appreciation is a facet of the broader positive body image construct, analogous to other facets such as body appreciation (Tylka & Wood-Barcalow, 2015) and functionality appreciation (Alleva et al., 2017).

To measure the construct of breast appreciation, Swami, Todd, and Tylka (2022) developed the 9-item Breast Appreciation Scale (BrAS). Based on data from samples of cisgender women in the United Kingdom, exploratory and confirmatory factor analyses supported the retention of a unidimensional model of the BrAS. This unidimensional model was found to be scalar invariant across mothers and non-mothers, as well across national groups (i.e., cisgender women from the UK, United States, and Australia). Swami, Todd, and Tylka (2022) also reported that the BrAS evidenced adequate 4-week test-retest reliability, as well as adequate convergent validity (e.g., positive associations with indices of positive body image), concurrent validity (e.g., positive associations with psychological well-being), discriminant validity (i.e., no significant association with impression management), and incremental validity (breast appreciation scores predicted unique variance in breast awareness above-and-beyond associations with breast size dissatisfaction).

To date, only two studies have assessed the psychometric properties of the BrAS outside English-speaking populations. Specifically, the 9-item unidimensional model of the BrAS was found to have adequate EFA- and CFA-based fit in cisgender samples of women from Romania (Swami et al., 2024) and Colombia (Mebarak et al., 2025). Both studies also reported that the unidimensional model of the BrAS achieved complete invariance (i.e., through to latent mean invariance) across mothers and non-mothers, had adequate composite reliability, and showed adequate evidence of convergent and concurrent validity. Additionally, Mebarak et al. (2025) also reported that BrAS scores remained temporally stable after four weeks. These results may be taken as evidence that the BrAS retains its robust psychometric properties in diverse national and linguistic contexts, but there remains a need to include further cultural and linguistic groups. This is important given that breasted experiences are likely shaped by a range of cultural and national factors, including patriarchal systems, cultural norms and values, idealised femininity and sexuality, heteronormativity, and the racialisation of beauty (Naugler, 2009; Swami et al., 2020).

### 1.1. The Lebanese context

To add to the emergent literature on breast appreciation in diverse cultural and linguistic groups, we sought to examine the psychometric properties of an Arabic translation of the BrAS in a sample of cisgender women from Lebanon. There are several reasons why this work may be of theoretical and practical interest to other scholars. First, development of a psychometrically valid Arabic translation of the BrAS would add to the growing inventory of instruments available to scholars in the Arabic-speaking world – primarily encompassing North Africa and the Middle East – for the assessment of indices of positive body image (e.g., Swami, Todd, Azzi et al., 2022; Swami et al., 2023). Indeed, despite recent advancements in the Middle East generally and in Lebanon specifically (e.g., Awad et al., 2024; Haddad et al., 2019), there remains a dearth of research on aspects of positive body image in Arabic-speaking populations. Second, Arab cultures and linguistics may shape breasted experiences in somewhat unique ways.

Historically, Arabic cultures exalted women's breasts as highly erotic (e.g., the breast features prominently in classical Arabic poetry; Drozdík, 1997). At the same time, however, popular epics and religious stories also venerated women's breasts for their role in child nurturance. In

many prophetic stories, for instance, mother figures are able to wield sizeable influence over their families and social worlds through their capacity to nurse (Schine, 2019). Over time, however, this coupling of the breasts with their functional purposes appears to have largely been lost or weakened across Arabic cultures. For instance, in their article-diary discussing motherhood and breast cancer, Alatrash and Zarif (2025) have noted how the Arabic word for "mother" (أم) is primarily used as a noun, with only rare instances of its use as a verb. Such linguistic limitations may entrench patriarchal norms that increasingly constrain the role and influence of motherhood in Arabic cultures (Ghazal, 2015), while simultaneously emphasising the aesthetic value of breasts.

The cultural shift toward emphasis on the aesthetic and symbolic value of the breast in Arabic cultures may also have roots elsewhere. For instance, some scholars have attributed this shift to the influence of European cultural imperialism, which sought to "orientalise" and objectify Arab women (see Mehdid, 1993). This has had a lasting legacy on the ways in which women in the Arab world navigate bodily experiences and hypersexual appearance ideals within broader socioeconomic and political contexts. In contemporary Lebanon, for instance, Melki et al. (2024) have discussed how cultural imperialism and heteronormative patriarchal systems have erased localised representations of women. For many young Lebanese women – irrespective of their communal and class identities – the breasts have become a site open to "investment" through breast augmentation for the sake of social and economic advancement (Doherty, 2008; Younes, 2019), a narrative fostered by the Lebanese state's neoliberal approach to body subjectivity (El Dine, 2025).

Beyond these issues, there is also a practical reason why development of an Arabic version of the BrAS may be useful. Specifically, Lebanese women have some of the highest rates of breast cancer incidence and mortality in the Middle East (Fares et al., 2019; Lakkis et al., 2010). Further, although the Lebanese Ministry of Public Health has facilitated access to mammography screenings for women aged 40 years and above across Lebanon since 2002 (Sbaity et al., 2011), there are several barriers to uptake, including poor knowledge and awareness of breast cancer (Doumit et al., 2017). As such, there may be value in promoting *breast awareness* (i.e., familiarity with one's breasts and the confidence to notice and detect changes to the breasts throughout a woman's life; McCready et al., 2005; Thornton & Pillarisetti, 2008) as a means of promoting improved breast cancer detection in Lebanese women. Given that breast appreciation is positively associated with breast awareness (Swami, Todd, & Tylka, 2022), it is likely that the availability of a validated measure of breast appreciation would benefit clinical practice in the context of Lebanon.

### 1.2. The present study

In view of the issues above, the present study sought to assess the psychometric properties of an Arabic translation of the BrAS in a sample of cisgender women from Lebanon. As in previous work (Mebarak et al., 2025; Swami, Todd, & Tylka, 2022; Swami et al., 2024), we used an EFA-to-CFA strategy, which allowed us to first establish the most suitable model of BrAS scores for our data without the imposition of modelling constraints (i.e., using EFA) and then to cross-validate and confirm the EFA-derived model (i.e., using CFA; Swami et al., 2021; Swami & Barron, 2019). We hypothesised that scores on the Lebanese version of the BrAS would reduce to a unidimensional model with all nine items retained, which would be consistent with previous work (Mebarak et al., 2025; Swami, Todd, & Tylka, 2022; Swami et al., 2024).

In addition, we also assessed construct validity of the BrAS in our sample. Selection of variables to assess the nomological network of the BrAS was guided by significant associations reported in previous work (Mebarak et al., 2025; Swami, Todd, & Tylka, 2022; Swami et al., 2024), relevant theoretical models of positive body image (e.g., Homan & Tylka, 2018), and the availability of instruments that have been

validated for use in Arabic-speaking populations. Specifically, to assess convergent validity, we examined associations between breast appreciation and indices of positive body image (body appreciation, functionality appreciation), symptoms of disordered eating, and breast awareness (operationalised using an instrument that we concurrently translated), with the expectation of small-to-moderate associations. To assess concurrent validity, we also examined associations with self-compassion and gratitude, with the expectation of small-to-moderate associations.

## 2. Method

### 2.1. Participants

The initial participant pool consisted of an online convenience sample of 656 cisgender women, but we excluded responses from 30 individuals who were missing > 80% of survey data. The final sample, therefore, consisted of 626 cisgender women who ranged in age from 18 to 50 years ( $M = 25.99$ ,  $SD = 9.47$ ,  $Mdn = 22.00$ , interquartile range = 8). Of the total sample, most were single (72.0 %), while smaller proportions were either married (22.5 %) or divorced (5.4 %). In terms of educational qualifications, 1.4 % had completed primary schooling, 15.0 % had completed secondary schooling, and 83.5 % had a university degree. No additional demographic data were collected.

### 2.2. Measures

#### 2.2.1. Breast appreciation

Participants were asked to complete a novel Arabic translation of the 9-item BrAS (Swami et al., 2022). To prepare this Arabic translation from its original in English, we used a combination of the back-translation method and a committee approach (Swami & Barron, 2019). First, two bilingual translators – one informed, and one uninformed – independently forward-translated the BrAS instructions, items, and response options into Arabic. Next, the two translations were examined by a third, independent bilingual translator who resolved any discrepancies and produce a synthesised translation. In a third step, the synthesised translation was back-translated by two bilingual translators naïve to the BrAS back into English. Finally, the forward- and back-translations were compared by an expert committee comprising all the translators, a methodologist, a linguist, two of the original developers of the BrAS, who resolved several minor inconsistencies between versions. We did not conduct a pilot test of understanding in the present case. The items of the BrAS in English and Arabic are presented in Table 1. Consistent with the original BrAS, all items were rated on a 5-point scale ranging from 1 (*strongly disagree*; Arabic: لا أوافق بشدة) to 5 (*strongly agree*; Arabic: أوافق بشدة). Higher item-level scores reflect greater breast appreciation.

#### 2.2.2. Body appreciation

The survey package also included the Body Appreciation Scale-2 (BAS-2; Tylka & Wood-Barcalow, 2015; Arabic translation: Vally

et al., 2019). This 10-item instrument assesses acceptance of one's body, respect and care for one's body, and protection of one's body from unrealistic beauty standards (sample item: "I respect my body"). All items were rated on a 5-point scale ranging from 1 (*never*) to 5 (*always*), with higher scores reflective of greater body appreciation. Scores on the Arabic version of the BAS-2 have been shown to have a unidimensional factor structure with adequate composite reliability and convergent validity (Swami et al., 2023), including in Lebanese women (Fekih-Romdhane et al., 2023). In the present study, composite reliability as assessed using McDonald's  $\omega$  for BAS-2 scores was .97 (95 %  $CI = .96, .98$ ).

#### 2.2.3. Functionality appreciation

Participants were asked to complete the Functionality Appreciation Scale (FAS; Alleva et al., 2017; Arabic translation: Swami et al., 2022). This 7-item instrument measures one's appreciation of what the body does and can do (sample item: "I appreciate my body for what it is capable of doing"). All items were rated on a 5-point scale, ranging from 1 (*strongly disagree*) to 5 (*strongly agree*), with higher scores reflecting greater functionality appreciation. Scores on the Arabic version of the FAS have been shown to have a unidimensional factor structure, adequate composite reliability, and adequate convergent and concurrent validity in Lebanese women (Swami et al., 2022). McDonald's  $\omega$  for FAS scores was .96 (95 %  $CI = .96, .97$ ).

#### 2.2.4. Body dissatisfaction

To measure body dissatisfaction, we used the Body Dissatisfaction subscale of the Eating Disorders Inventory-3 (EDI-BD; Garner, 2004; Arabic translation: Haddad et al., 2019). This is a 9-item instrument that measures dissatisfaction with various body parts (sample item: "I think that my stomach is too big"), with items were rated on a 5-point scale (1 = *never*, 5 = *always*). Although nominally measuring a singular underlying construct (Garner, 2004), factor analytic results with Lebanese adults has shown that scores on the subscale reduce to two dimensions (Gerges et al., 2023), likely an artefact of including positive and negatively valenced items in the same instrument. As such, we computed an index of body dissatisfaction using the five negatively valenced items only, with higher scores reflective of greater body dissatisfaction. McDonald's  $\omega$  for EDI-BD scores was .85 (95 %  $CI = .83, .87$ ).

#### 2.2.5. Symptoms of disordered eating

Participants were also asked to complete a short-form, 7-item version of the Eating Attitudes Test (Garner et al., 1982) in Arabic (EAT-7; Fekih-Romdhane et al., 2022). This short form of the instrument assesses symptoms and concerns characteristic of food- and eating-related disordered behaviours (sample item: "I avoid eating when I am hungry"). All items were rated on a 6-point scale ranging from 1 (*never*) to 6 (*always*), with higher scores reflective of characteristics of disordered eating. Scores on the EAT-7 have been shown to have a unidimensional factor structure and adequate convergent validity in Lebanese women (Fekih-Romdhane et al., 2022). In the present study, McDonald's  $\omega$  for EAT-7 scores was .87 (95 %  $CI = .86, .89$ ).

**Table 1**

Items of the Breast Appreciation Scale (BrAS) in English and Arabic, and Factor Loadings Derived from the Exploratory Factor Analyses (EFA) in the First Split-Half Subsample, and Standardised Estimates of Factor Loadings from the Confirmatory Factor Analysis (CFA) in the Second Split-Half Subsample.

| English                                                                                              | Arabic                                                                                     | EFA | CFA |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----|-----|
| I take a positive attitude toward my breasts.                                                        | أأخذ موقفاً إيجابياً تجاه ثدياتي.                                                          | .87 | .79 |
| I feel good about how my breasts look.                                                               | أشعر بالرضا عن شكل ثدياتي.                                                                 | .86 | .72 |
| I am grateful for all that my breasts allow me to do (e.g., experience pleasure, breastfeed).        | (أنا ممتنة لكل ما يسمح لي ثدياتي بفعله (على سبيل المثال، تجربة المتعة، والرضاعة الطبيعية). | .66 | .68 |
| I am comfortable with the way my breasts have changed over time.                                     | أشعر بالارتياح تجاه الطريقة التي تغير بها ثدياتي مع مرور الوقت.                            | .83 | .80 |
| I feel love for my breasts.                                                                          | أشعر بالحب تجاه ثدياتي.                                                                    | .87 | .86 |
| I am happy with how my breasts feel.                                                                 | أشعر بالسعادة تجاه كيف أشعر بثدياتي.                                                       | .87 | .83 |
| I appreciate the different and unique characteristics of my breasts.                                 | أقدر الخصائص المختلفة والفريدة من نوعها لثدياتي.                                           | .78 | .79 |
| All things considered, I am satisfied with my breasts.                                               | في كل الأحوال، أنا راضية عن ثدياتي.                                                        | .88 | .82 |
| I think my breasts are beautiful, even if they are different from media images of idealized breasts. | أعتقد أن ثدياتي جميلان، حتى لو كانا مختلفين عن الصور الإعلامية للثدي المثالي.              | .83 | .83 |

### 2.2.6. Gratitude

To measure the construct of gratitude, participants were asked to complete the Gratitude Questionnaire-6 (GQ-6; McCullough et al., 2002; Arabic translation: Fekih-Romdhane et al., 2025). In its original version, this instrument consists of 6 items assessing one's disposition toward gratitude (sample item: "I have so much in life to be thankful for"). In Arabic, however, only a 4-item version that excludes two negatively worded items has been shown to have adequate factor analytic fit (GQ-4; Fekih-Romdhane et al., 2025). As such, we computed gratitude using this 4-item version, with higher scores reflective of greater gratitude. All items were rated on a 7-point scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The 4-item version of the Gratitude Questionnaire has been shown to have adequate concurrent validity (Fekih-Romdhane et al., 2025). In the present study, McDonald's  $\omega$  for this instrument was .95 (95 % CI = .95, .96).

### 2.2.7. Self-compassion

Self-compassion was assessed using the 12-item Self-Compassion Scale-Short Form (SCS-SF; Raes et al., 2011; Arabic translation: Swami et al., 2025). The SCS-SF uses two items per component to measure six hypothesised components of self-compassion, namely self-kindness, mindfulness, common humanity, self-judgement, overidentification, and isolation (sample item: "When something upsets me, I try to keep my emotions in balance"). All items were rated on a 5-point scale, ranging from 1 (*almost never*) to 5 (*almost always*). Recent work, including with Lebanese adults, has shown that scores on the scale are best conceptualised as two distinct factors assessing compassionate (five items) and uncompassionate self-responding (6 items), respectively (Swami et al., 2025). Thus, for the present purposes, we computed an index of compassionate self-responding using the five relevant items. Higher scores are, therefore, reflective of greater compassionate self-responding. This variant of the SCS-SF has been shown to have adequate psychometric properties in diverse samples, including Lebanese adults (Swami et al., 2025). In the present study, McDonald's  $\omega$  for this instrument was .88 (95 % CI = .86, .89).

### 2.2.8. Breast awareness

To measure breast awareness, we used the measure of breast awareness developed by Swami, Todd, and Tylka (2022). This 3-item instrument assesses familiarity with one's breasts and the confidence to notice and detect changes to the breasts throughout one's life (sample item: "I have a good idea of how my breasts will change over my lifetime"). An Arabic version of this instrument was developed using the same 5-step translational strategy as described above (see [Supplementary Table S1](#) for items in Arabic and English). In the main survey, all items were rated on a 5-point scale (1 = *strongly disagree*, 5 = *strongly agree*). Because this instrument has not been previously used in Arabic, we subjected our data to an EFA to assess its factor structure. Our results showed that it was possible to extract a single factor with all three items, common variance among the items was deflated (indicated by a low Kaiser-Meyer-Olkin value; see [Supplementary Materials](#)). On this basis, we computed an index of breast awareness using all three items, with higher scores reflecting greater breast awareness. In the present study, McDonald's  $\omega$  for this instrument was .78 (95 % CI = .74, .81).

### 2.2.9. Demographics

Participants were asked to provide their demographic details consisting of age, highest educational qualification, and marital status.

## 2.3. Procedures

Once ethics approval for this study was obtained from the Hospital of Rayak (approval code: ECO-R-616), the study was advertised on social media. Inclusion criteria for participation including self-identifying as a cisgender woman, a resident and citizen of Lebanon, being of adult age, and fluent in Arabic. We also excluded women who were pregnant at the

time of the study, who had a history of breast cancer, and who had breast augmentation surgery. Potential participants who were eligible for inclusion provided digital informed consent before completing an online survey hosted on Google Forms. All data were collected between March and August 2025, and the order of presentation of the instruments described above was pre-randomised for each participant. To support the collection of high-quality data, we used CAPTCHA verification to prevent responses from bots and restricted survey submissions to one per IP address (to avoid duplicate entries). The survey was anonymous and participants completed the survey voluntarily and without remuneration. All participants received debriefing information at the end of the survey.

## 2.4. Analytic strategy

### 2.4.1. Data treatment

There were no missing data in the retained dataset. To examine the factor structure of the BrAS, we used an EFA-to-CFA strategy as recommended by Swami and Barron (2019). To ensure adequate sample sizes for both EFA and CFA, we split the sample using a computer-generated random seed, resulting in one split-half for EFA ( $n = 312$ ) and a second split-half for CFA ( $n = 314$ ). There were no significant differences across the two split-halves in terms of mean age,  $t(624) = 0.61$ ,  $p = .544$ ,  $d = .05$ , as well as the distribution of marital status,  $\chi(2) = 6.14$ ,  $p = .058$ , and education qualifications,  $\chi(2) = 1.47$ ,  $p = .689$ .

### 2.4.2. Exploratory factor analysis

Data from the first split-half were subjected to principal-axis EFA using the *psych* package (Revelle, 2019) in R (R Development Core Team, 2014). Our subsample size satisfied [Worthington and Whittaker's \(2006\)](#) item-communality requirements, as well as assumptions for EFA based on item distributions, average item correlations, and item-total correlations (Clark & Watson, 1995). As per [Hair et al. \(2009\)](#), data factorability was assessed using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy (which should ideally be  $\geq .80$ ) and Bartlett's test of sphericity (which should be significant). Principal-axis factoring was used for the EFAs as it yields results similar to commonly used maximum likelihood estimation without assuming multivariate normality ([Goretzko et al., 2021](#)). Given the expectation of a single orthogonal factor, a quartimax rotation was applied ([Pedhazur & Schmelkin, 1991](#)).

To estimate the number of factors to extract and factor structure adequacy, we considered the results of a parallel analysis test ([Horn, 1965](#)) and examined fit statistics using commonly used fit indices ([Finch, 2020](#); [Swami & Barron, 2019](#)). Specifically, we used the normed model chi-square ( $\chi^2/df$ ; values  $< 3.0$  considered indicative of good fit), the Steiger-Lind root mean square error of approximation (RMSEA) and its 90 % CI (values close to .06 considered to be indicative of good fit and up to about .08 indicative of adequate fit), the standardised root mean square residual (SRMR; values  $< .09$  indicative of good fit), the Tucker-Lewis index (TLI; values close to or  $> .95$  indicative of good fit), and the comparative fit index (CFI; values close to or  $> .95$  indicative of adequate fit) ([Hu & Bentler, 1999](#)). Corrections to fit indices were not required as EFA is robust to violations of univariate and multivariate normality ([Curran et al., 1996](#)). Item retention was based on the recommendation that items with "fair" loadings and above (i.e.,  $\geq .33$ ) and with low inter-item correlations (suggestive of low item redundancy) as indicated by the anti-image correlation matrix should be retained ([Comrey & Lee, 1992](#)).

### 2.4.3. Confirmatory factor analysis

We used data from the second split-half to conduct a CFA using the *lavaan* (Rosseel, 2012), *semTools* (Jorgensen et al., 2018), and *MVN* packages ([Korkmaz et al., 2014](#)) with R (R Development Core Team, 2014). Based on proactive Monte Carlo simulations, [Swami et al. \(2024\)](#) previously reported that a sample size of about 160 would be sufficient

for this analysis, which was surpassed in our subsample. Our aim was to test the unidimensional model proposed by Swami, Todd, and Tylka (2022) and, if discrepant, the model suggested by our EFA results. Assessment of the data for univariate normality indicated acceptable levels of skewness (values ranged from  $-0.77$  to  $-0.35$ ) and kurtosis (values ranged from  $-0.12$  to  $-0.32$ ), but assessment of multivariate normality indicated that the data were not multivariate normal (Mardia's skewness =  $1208.38$ ,  $p < .001$ , Mardia's kurtosis =  $59.48$ ,  $p < .001$ ). Therefore, parameter estimates were obtained using the robust maximum likelihood method and fit indices (see Section 2.4.2) were interpreted with the Satorra-Bentler correction applied (Satorra & Bentler, 2001).

2.4.4. Composite reliability

Composite reliability in all analyses was assessed using McDonald's  $\omega$  and its associated 95 % CI, with values greater than .70 reflecting adequate internal reliability (Dunn et al., 2014). Specifically, we computed  $\omega$  using the *semTools* package for R (Jorgensen et al., 2018), which estimates reliability based on CFA parameters and allows for models that do not fit the data perfectly (Kelley & Pornprasertmanit, 2016).

2.4.5. Construct validity

With data from the second split-half subsample, we assessed discriminant validity using the Fornell-Larcker criterion (Fornell & Larcker, 1981), with average variance extracted (AVE) values of  $\geq .50$  considered adequate and meaning that a latent variable is able to explain more than half of the variance of its indicators on average (i.e., items converge into a uniform construct). To assess construct validity (i.e., convergent and concurrent validity), we used the total sample to construct a structural equation model (SEM) in which the BrAS factor structure was estimated based on model providing the most optimal representation of the data across both split-half subsamples. In this model, the latent factors of breast appreciation, body appreciation, functionality appreciation, body dissatisfaction, disordered eating symptoms, gratitude, compassionate self-responding, and breast awareness were all correlated. We also report bivariate Spearman correlations with manifest (observed) scores for completeness. Based on Cohen (1992), values  $\leq .10$  were considered weak,  $\sim .30$  were considered moderate, and  $\sim .50$  were considered strong correlations.

3. Results

3.1. Exploratory factor analysis

Both Bartlett's test of sphericity,  $\chi^2(36) = 2589.90$ ,  $p < .001$ , and the KMO measure of sampling adequacy (.94), indicated that the BrAS items had adequate common variance for factor analysis in the first split-half subsample. Principal-axis EFA indicated that a single factor, which explained 69.1 % of common variance, should be extracted. This was confirmed by the parallel analysis test ( $\lambda_1 = 6.23$ ;  $\lambda_2 = 0.58 < 0.98$ ). The fit indices for this model were:  $\chi^2(27) = 180.32$ ,  $p < .001$ , CFI = .941, TLI = .921, RMSEA = .135 (90 % CI = .117, .154), SRMR = .033. Factor loadings, reported in Table 1, indicated that all nine items should be retained. In this subsample, BrAS scores had adequate composite reliability ( $\omega = .95$ , 95 % CI = .94, .96).

3.2. Confirmatory factor analysis

CFA fit indices indicated that fit of the unidimensional model of BrAS scores was acceptable:  $SB\chi^2(27) = 67.32$ ,  $p < .001$ ,  $SB\chi^2_{normed} = 2.49$ , robust RMSEA = .069 (90 % CI = .056, .082), SRMR = .043, robust CFI = .949, robust TLI = .932. The standardised estimates of factor loadings were all adequate (see Table 1). In this subsample, BrAS scores had adequate composite reliability ( $\omega = .94$ , 95 % CI = .92, .95).

3.3. Construct Validity

Discriminant validity for the unidimensional model of BrAS was adequate, as AVE = .63. In terms of convergent and concurrent validity, evaluation of the goodness-of-fit statistics for the SEM including latent scores for all measures included in the survey indicated acceptable model fit in the total sample,  $\chi^2(1196) = 2098.02$ ,  $p < .001$ , robust CFI = .953, TLI = .950, robust RMSEA = .041 (90 % CI = .038, .044), SRMR = .045. As reported in Table 2, latent breast appreciation scores had moderate-to-strong positive correlations with latent scores for breast awareness, body appreciation, functionality appreciation, gratitude, and compassionate self-responding, and small-to-moderate negative associations with body dissatisfaction and symptoms of disordered eating. The same pattern of results was observed when examining associations with manifest scores, which were based on the computation of mean values (see Table 2).

4. Discussion

In the present study, our objective was to assess the psychometric properties of a novel Arabic translation of the BrAS (Swami, Todd, & Tylka, 2022) in a sample of Lebanese cisgender women. In terms of factorial validity, we adopted an EFA-to-CFA analytic strategy, which showed that it was possible to retain a unidimensional model of the BrAS. This model included all nine of the original BrAS items and showed adequate composite reliabilities across both split-half subsamples. Our results also supported the construct (i.e., convergent and concurrent) validity of the Arabic BrAS, insofar as (latent and manifest) associations with additional constructs were in the expected directions. In broad outline, these results are consistent with earlier test adaptation studies, which have shown the BrAS has adequate psychometric properties in Romanian-speaking women from Romania (Swami et al., 2024) and Spanish-speaking women from Colombia (Mebarak et al., 2025).

Turning first to factorial validity, the results of our EFA and parallel analysis in the first split-half subsample suggested that a single BrAS factor should be extracted. This unidimensional model included all nine original BrAS items with item-factor loadings that were consistently good. In addition, our cross-validation with CFA in the second split-half subsample showed that the unidimensional model of the BrAS has adequate fit to our data, with standardised estimates of factor loadings again consistently good. Additionally, composite reliability was adequate in both split-half subsamples. Taken together, these results suggests that the Arabic BrAS can be reliably considered to be represented by a unidimensional model consisting of all nine items. This is consistent with the results of the parent development study (Swami, Todd, & Tylka, 2022), as well as previous test adaptation work (Mebarak et al., 2025; Swami et al., 2024).

In the present study, we also examined the construct validity of the Arabic BrAS. In terms of convergent validity, we found that latent BrAS scores were moderately and positively associated with latent body appreciation and functionality appreciation. This is consistent with the claim that breast appreciation can be considered to be a facet of the

**Table 2**  
Associations between Breast Appreciation (Latent and Observed Scores) and All Other Variables Included in the Study.

|                               | Latent Score      | Observed Score ( $r_s$ ) |
|-------------------------------|-------------------|--------------------------|
| Body appreciation             | .60               | .51                      |
| Functionality appreciation    | .58               | .50                      |
| Body dissatisfaction          | -.40              | -.33                     |
| Symptoms of disordered eating | -.08 <sup>a</sup> | -.11 <sup>b</sup>        |
| Gratitude                     | .47               | .39                      |
| Compassionate self-responding | .29               | .24                      |
| Breast awareness              | .65               | .50                      |

Note.  $r_s$  = Spearman correlations, all  $p$  values  $< .001$  except the following: <sup>a</sup> = .087, <sup>b</sup> = .008.

broader positive body image construct (Swami, Todd, & Tylka, 2022). Additional support for convergent validity came from the finding that breast appreciation was moderately and negatively associated with body dissatisfaction and symptoms of disordered eating. In terms of concurrent validity, we found that latent breast appreciation was positively and moderately associated with gratitude and compassionate self-responding. In broad outline, these findings are consistent with evidence of convergent and concurrent validity of the BrAS presented elsewhere (Mebarak et al., 2025; Swami, Todd, & Tylka, 2022; Swami et al., 2024).

Finally, and again supporting the convergent validity of the BrAS, we found that a moderate and positive association between latent breast appreciation and breast awareness. This is an important finding in its own right because, as we noted earlier, Lebanese women have some of the highest rates of breast cancer incidence and mortality in the Middle East (Fares et al., 2019; Lakkis et al., 2010). One way of potentially reducing the incidence rate of breast cancer is through the promotion of breast awareness (McCready et al., 2005; Thornton & Pillarisetti, 2008). In view of the positive association between these constructs in the present study, it might be suggested that Lebanese women may benefit in terms of breast cancer detection from the promotion of greater breast appreciation and its downstream impact on breast awareness (Swami, Todd, & Tylka, 2022), although of course any causal conclusions should be treated with caution.

Of note, however, the measure of breast awareness used in the present study – which we obtained from Swami, Todd, and Tylka (2022) – showed evidence of deflated common variance among the three items. This is consistent with previous work (Mebarak et al., 2025; Swami et al., 2024) and suggests that this instrument may be assessing disparate components of a broader breast awareness construct. While we have used this instrument with caution in the present study, we recommend that scholars consider revising this instrument or developing an improved measure of breast awareness in the future. This would help to provide greater confidence that the association between breast appreciation and breast awareness, or components of breast awareness, is indeed reliable and valid. Meanwhile, the results presented in [Supplementary Materials](#) suggest that the existing measure of breast awareness may provide a unidimensional assessment of the construct in Lebanese women.

There are several other limitations of the present study, which should be addressed in future work. First, we did not include a measure of breast dissatisfaction (e.g., Swami et al., 2015), which would have provided an additional assessment of convergent validity and would have allowed us to test for incremental validity (Mebarak et al., 2025). In future work, it may also be useful to examine indices of discriminant validity, as well as temporal stability of BrAS scores. In a similar vein, we also neglected to ask participants about any infant feeding practices and motherhood status (i.e., the number of children they had given birth to). Swami, Tylka, and Todd (2022) suggested that higher breast appreciation in mothers may be taken as evidence of known-groups validity, but more recent work has suggested that there are no significant differences between mothers and non-mothers in latent breast appreciation (Mebarak et al., 2025; Swami et al., 2024). Nevertheless, this may be an interesting aspect for future research, particularly given the pressure that Lebanese women experience to become mothers and the anxiety they experience when they remain childless (Kaak et al., 2025).

In future work, it may also be important to consider the ways that demographic diversity affect breast appreciation. For instance, scholars have discussed how religious identities, class structures, and neoliberalism have intersected to exaggerate communal identities that may have an impact on body image among women in Lebanon (Deeb, 2006; El Dine, 2025). As such, it may be useful to consider whether there are differences in breast appreciation between Lebanese Christians and Sunni Muslims (who have historically held greater political power) and Shia Muslims (who have historically been politically marginalised and tend to be more conservative in terms of religion). In a similar vein, it

may also be useful to consider the role that food insecurity plays in shaping breast appreciation, particularly given evidence that Lebanese women are disproportionately burdened by food deprivation (e.g., likelihood of engaging in self-sacrificial behaviour) and its impacts on psychological well-being (Mattar et al., 2023).

Finally, there are some more generic limitations that should be considered when assessing our findings. For example, our recruitment method means that our sample is unlikely to have been representative of the wider Lebanese population of women. Indeed, our sample was relatively young, highly educated, and predominantly single, which may mean that their experiences of breast appreciation have limited generalisability in the Lebanese context. We also did not request residential information and, given our reliance on online recruitment, we cannot entirely rule out geographic clustering (e.g., recruitment being concentrated in particular locales), which may have affected responding on our survey. In a similar vein, the focus of our work was on cisgender women and it may be useful in future work to investigate the applicability of the BrAS to understanding breast experiences in populations of other gender identities (e.g., transwomen and people assigned as female at birth with breasts but who not identify as women). Finally, it should not be assumed that our findings will generalise to other Arabic-speaking populations. Rather, the psychometric properties of the Arabic BrAS should ideally be tested any time it is deployed in a new national context (Swami & Barron, 2019).

These limitations notwithstanding, the results of the present study suggest that the BrAS is a reliable and valid tool for the assessment of breast appreciation in Lebanese women. The availability of this translation of the BrAS adds to the tools available to researchers in Lebanon who are interested in issues related to positive body image, which in turn may spur more in-depth understandings of this construct in the context of the Middle East and Arabic-speaking communities. For body image scholars more broadly, the present results suggest that it may be possible to capture a common latent construct of breast appreciation across linguistic and national groups using the BrAS. This raises interesting questions about the extent to which the BrAS will evidence measurement invariance, which could be assessed in future cross-national research. Such research can be expected to contribute to more holistic understandings of the positive body image construct, as well as fuller consideration of positive breast experiences across cultural communities.

#### CRedit authorship contribution statement

**Viren Swami:** Writing – original draft, Validation, Methodology, Conceptualization. **Souheil Hallit:** Project administration, Investigation, Data curation, Conceptualization. **Jennifer Todd:** Writing – review & editing, Validation. **Feten Fekih-Romdhane:** Writing – review & editing, Investigation. **Sahar Obaid:** Writing – review & editing, Investigation. **Mirna Fawaz:** Writing – review & editing, Investigation. **Nour El Achkar:** Writing – review & editing, Investigation. **Diana Malaeb:** Writing – review & editing, Investigation.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.bodyim.2026.102031](https://doi.org/10.1016/j.bodyim.2026.102031).

#### Data availability

Data will be made available on request.

## References

- Alatrash, G., & Zarif, M. (2025). Diary of losing a breast and reflections on mothering as an Arab. *Journal of the Motherhood Initiative for Research and Community Involvement*, 15, 17.
- Alleva, J. M., Tylka, T. L., & Kroon Van Diest, A. M. (2017). The Functionality Appreciation Scale (FAS): Development and psychometric properties in U.S. community women and men. *Body Image*, 23, 28–44. <https://doi.org/10.1016/j.bodyim.2017.07.008>
- Awad, E., Zaklat, N., Rahme, E., Hallit, S., Fekih-Romdhane, F., & Obeid, S. (2024). Body appreciation and mindful eating in Lebanese adults: The mediating role of psychological distress. *International Journal of Environmental Health Research*, 34, 4020–4032. <https://doi.org/10.1080/096031123.2024.2337832>
- Clark, L. A., & Watson, D. (1995). Constructing validity: Basic issues in objective scale development. *Psychological Assessment*, 7(3), 309–319. <https://doi.org/10.1037/1040-3590.7.3.309>
- Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112(1), 155–159. <https://doi.org/10.1037/0033-2909.112.1.155>
- Comrey, A. L., & Lee, H. B. (1992). *A first course in factor analysis*. Lawrence Erlbaum.
- Curran, P. J., West, S. G., & Finch, J. F. (1996). The robustness of test statistics to nonnormality and specification error in confirmatory factor analysis. *Psychological Methods*, 1(1), 16–29. <https://doi.org/10.1037/1082-989X.1.1.16>
- Daniels, E. A., Gillen, M. M., & Markey, C. H. (Eds.). (2018). *Body positive: Understanding and improving body image in science and practice*. Cambridge University Press.
- Deeb, L. (2006). *An enchanted modern: Gender and public piety in Shi'i Lebanon*. Princeton University Press. <https://doi.org/10.1515/9781400840786>
- Doherty, S. B. (2008). Cosmetic surgery and the beauty regime in Lebanon. *Middle East Report*, 249, 28–31.
- Doumit, M. A. A., Fares, S., & Arevian, M. (2017). Knowledge, practices, and attitudes of women towards breast cancer in Lebanon. *American Journal of Hematology/Oncology*, 13(7), 21–28.
- Drozdzik, L. (1997). Erotic imagery in classic Arabic poetry. *Asian and African Studies*, 6, 21–44.
- Dunn, T. J., Baguley, T., & Brunsden, V. (2014). From alpha to omega: A practical solution to the pervasive problem of internal consistency estimation. *British Journal of Psychology*, 105(3), 399–412. <https://doi.org/10.1111/bjop.12046>
- El Dine, S. N. (2025). Imagining sect and class through cosmetic surgery: 'Self-care' as social distinction in Lebanon. *Middle East Critique Advance Online Publication*. <https://doi.org/10.1080/19436149.2025.2499761>
- Fares, M. Y., Salhab, H. A., Khachfe, H. H., & Khachfe, H. M. (2019). Breast cancer epidemiology among Lebanese women: An 11-year analysis. *Medicina*, 55(8), 463. <https://doi.org/10.3390/medicina55080463>
- Fekih-Romdhane, F., Azzi, V., Malaeb, D., El Dine, A. S., Obeid, S., & Hallit, S. (2023). Psychometric properties of an Arabic translation of the Body Appreciation Scale-2 (BAS-2) and its short forms (BAS-2SF) in a community sample of Lebanese adults. *Journal of Eating Disorders*, 11, 160. <https://doi.org/10.1186/s40337-023-00885-x>
- Fekih-Romdhane, F., Obeid, S., Malaeb, D., Hallit, R., & Hallit, S. (2022). Validation of a shortened version of the Eating Attitude Test (EAT-7) in the Arabic language. *Journal of Eating Disorders*, 10, 127. <https://doi.org/10.1186/s40337-022-00651-5>
- Fekih-Romdhane, F., Postigo, Á., Malaeb, D., El Dine, A. S., Moawad, M., Hallit, R., & Hallit, S. (2025). Validation of the Arabic version of the Gratitude Questionnaire (GQ-4) in a sample of non-clinical adults. *BMC Psychology*, 13, 143. <https://doi.org/10.1186/s40359/025-02441-5>
- Finch, W. H. (2020). Using fit statistic differences to determine the optimal number of factors to retain in exploratory factor analysis. *Educational and Psychological Measurement*, 80(2), 217–241. <https://doi.org/10.1177/0013164419865769>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://doi.org/10.2307/3151312>
- Garner, D. M. (2004). *Eating Disorder Inventory-3: Professional manual*. Psychological Assessment Resources.
- Gerald, W., & Potvin, L. (2009). Boobs, boxing, and bombs: Problematizing the entertainment of Spike TV. *Spaces for Difference*, 2(1), 3–14.
- Gerges, S., Obeid, S., Malaeb, D., El Dine, A. S., Hallit, R., Soufiah, M., Fekih-Romdhane, F., & Hallit, S. (2023). Validation of an Arabic version of the Eating Disorder Inventory's Body Dissatisfaction subscale among adolescents, adults, and pregnant women. *Journal of Eating Disorders*, 11, 187. <https://doi.org/10.1186/s40337-023-00911-y>
- Ghazal, A. (2015). *Motherhood in Arabic/African literary texts* (Unpublished Masters Thesis). Beirut, Lebanon: Lebanese American University.
- Goretzko, D., Pham, T. T. H., & Bühner, M. (2021). Exploratory factor analysis: Current use, methodological developments and recommendations for good practice. *Current Psychology*, 40, 3510–3521. <https://doi.org/10.1007/s12144-019-00300-2>
- Haddad, C., Zakhour, M., Akel, M., Honein, K., Akiki, M., Hallit, S., & Obeid, S. (2019). Factors associated with body dissatisfaction among the Lebanese population. *Eating and Weight Disorders*, 24, 507–519. <https://doi.org/10.1007/s40519-018-00634-z>
- Hair, J. F., Jr., Black, W. C., Babin, B. J., & Anderson, R. E. (2009). *Multivariate data analysis* (7th ed.). Macmillan.
- Helm, R. L., O'Hea, E. L., & Corso, E. L. (2008). Body image issues in women with breast cancer. *Psychology, Health Medicine*, 13(3), 313–325. <https://doi.org/10.1080/13548500701405509>
- Homan, K. J., & Tylka, T. L. (2018). Development and exploration of the gratitude model of body appreciation in women. *Body Image*, 25, 14–22. <https://doi.org/10.1016/j.bodyim.2018.01.008>
- Horn, J. L. (1965). A rationale and test for the number of factors in factor analysis. *Psychometrika*, 32, 179–185. <https://doi.org/10.1007/BF02289447>
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- Jorgensen, T. D., Pornprasertmanit, S., Schoemann, A. M., & Rosseel, Y. (2018). *semTools: Useful tools for structural equation modeling*. R package version 0.5-1. (<https://CRAN.R-project.org/package=semTools>).
- Kaak, S., Ziade, F., & Mansour, C. (2025). Psychological impact of menopause on childless women in Lebanon: A comparative study of self-esteem, anxiety, and body image. *Rihan Journal for Science Publishing*, 55, 144.
- Kelley, K., & Pornprasertmanit, S. (2016). Confidence intervals for population reliability coefficients: Evaluation of methods, recommendations, and software for composite measures. *Psychological Methods*, 21(1), 69–92. <https://doi.org/10.1037/a0040086>
- Korkmaz, S., Goksuluk, D., & Zararsiz, G. (2014). MVN: An R package for assessing multivariate normality. *The R Journal*, 6, 151–162.
- Lakkis, N. A., Adib, S. M., Osman, M. H., Musharrafieh, U. M., & Hamadeh, G. N. (2010). Breast cancer in Lebanon: Incidence and comparison to regional and Western countries. *Cancer Epidemiology*, 34(3), 221–225. <https://doi.org/10.1016/j.canep.2010.02.013>
- Lee, J. (1997). Never innocent: Breast experiences in women's bodily narratives of puberty. *Feminism Psychology*, 7(4), 453–474. <https://doi.org/10.1177/0959353597074002>
- Lombardo, C., Panasiti, M. S., Vacca, M., Grano, C., & Swami, V. (2019). Motherhood status moderates the relationship between perfectionistic self-presentation and breast size satisfaction. *Body Image*, 30, 75–80. <https://doi.org/10.1080/j.bodyim.2019.05.005>
- Mattar, L., Mashharawi, F., Harb, H., & El Khoury, M. (2023). Food insecurity in Lebanon: Impact on maternal mental health and coping mechanisms. *Clinical Nutrition ESPEN*, 42e750, 58.
- McCann, H. (2020). 'The free-flying natural woman boobs of yore'? The body beyond representation in feminist accounts of objectification. *Feminist Review*, 126(1), 74–88. <https://doi.org/10.1177/0141778920944550>
- McCready, T., Littlewood, D., & Jenkinson, J. (2005). Breast self-examination and breast awareness: A literature review. *Journal of Clinical Nursing*, 14(5), 570–578. <https://doi.org/10.1111/j.1365-2702.2004.01108.x>
- Mebarak, M., Mendoza, J., Todd, J., Chamorro Coneo, A. M., Muñoz-Alvis, A., de los Reyes Aragon, C., Herrera, M., Trejos Herrera, A. M., Martínez-Banfi, M., Polo-Vargas, J. D., Díaz-Granados, E. A., Gutierrez Carvajal, O. I., Carrasquilla, A., Mercado Castro, S., & Swami, V. (2025). Translation and validation of a (Colombian) Spanish version of the Breast Appreciation Scale (BrAS). *Body Image*, 53, Article 101888. <https://doi.org/10.1016/j.bodyim.2025.101888>
- Mehdi, M. (1993). A Western invention of Arab womanhood: The 'Oriental' female. In H. Afshar (Ed.), *Women in the Middle East: Women's Studies at York Series* (pp. 18–58). London: Palgrave Macmillan. [https://doi.org/10.1007/9781-349-22588-0\\_2](https://doi.org/10.1007/9781-349-22588-0_2)
- Melki, J., King, G., Sakka, Y., Haidar, R., Saati, F., & Mazloun, N. (2024). Gendered representations in Lebanese advertisements: Flawless white femininity, heteronormative patriarchal masculinity, and the erasure of local diversity. *Feminist Media Studies*, 24(2), 240–257. <https://doi.org/10.1080/14680777.2023.2194571>
- Millsted, R., & Frith, H. (2003). Being large-breasted: Women negotiating embodiment. *Women's Studies International Forum*, 26(5), 455–465. <https://doi.org/10.1016/j.wsif.2003.08.003>
- Nash, M. (2014). Breast experience in pregnancy: An examination through photographs. *Visual Studies*, 29(1), 40–53. <https://doi.org/10.1080/1472586X.2014.862992>
- Pedhazur, E. J., & Schmelkin, L. P. (1991). *Measurement, design, and analysis: An integrated approach*. Lawrence Erlbaum.
- R Development Core Team. (2014). R: A language and environment for statistical computing. R Foundation for Statistical Computing.
- Raes, F., Pommier, E., Neff, K. D., & van Gucht, D. (2011). Construction and factorial validation of a short form of the Self-Compassion Scale. *Clinical Psychology and Psychotherapy*, 18(3), 250–255. <https://doi.org/10.1002/cpp.702>
- Revelle, W. R. (2019). *psych: Procedures for personality and psychological research*. (<https://cran.r-project.org/web/packages/psych/index.html>).
- Rosseel, Y. (2012). *lavaan: An R package for structural equation modeling*. *Journal of Statistical Software*, 48, 1–36.
- Satorra, A., & Bentler, P. M. (2001). A scaled difference chi-square test statistic for moment structure analysis. *Psychometrika*, 66, 507–514. <https://doi.org/10.1007/BF02296192>
- Sbaity, E., Bejjany, R., Kreidieh, M., Temraz, S., & Shamseddine, A. (2011). Overview in breast cancer screening in Lebanon. *Cancer Control*, 28, 1. <https://doi.org/10.1177/10732748211039443>
- Schmied, V., & Barclay, L. (1999). Connection and pleasure, disruption and distress: Women's experiences of breastfeeding. *Journal of Human Lactation*, 15(4), 325–334. <https://doi.org/10.1177/089033449901500410>
- Schine, R. (2019). Nourishing the noble: Breastfeeding and hero-making in Medieval Arabic popular literature. *Al-'Usir al-Wustā*, 27, 165–200. <https://doi.org/10.7916/alusur.v27il.6784>
- Swami, V., & Barron, D. (2019). Translation and validation of body image instruments: Challenges, good practice guidelines, and reporting recommendations for test adaptation. *Body Image*, 31, 204–220. <https://doi.org/10.1016/j.bodyim.2018.08.014>
- Swami, V., Cavelti, S., Taylor, D., & Tovée, M. J. (2015). The Breast Size Rating Scale: Development and psychometric evaluation. *Body Image*, 14, 29–38. <https://doi.org/10.1016/j.bodyim.2015.02.004>
- Swami, V., & Furnham, A. (2018). Breast size dissatisfaction, but not body dissatisfaction, is associated with breast self-examination frequency and breast

- change behaviours in British women. *Body Image*, 24, 76–81. <https://doi.org/10.1016/j.bodyim.2017.12.004>
- Swami, V., Cavelti, S., Taylor, D., & Tovée, M. J. (2015). The Breast Size Rating Scale: Development and psychometric evaluation. *Body Image*, 14, 29–38. <https://doi.org/10.1016/j.bodyim.2015.02.004>
- Swami, V., Todd, J., & Barron, D. (2021). Translation and validation of body image instruments: An addendum to Swami and Barron (2019) in the form of frequently asked questions. *Body Image*, 37, 214–224. <https://doi.org/10.1016/j.bodyim.2021.03.002>
- Swami, V., Todd, J., Azzi, V., Malaeb, D., El Dine, A. S., Obeid, S., & Hallit, S. (2022). Psychometric properties of an Arabic translation of the Functionality Appreciation Scale in Lebanese adults. *Body Image*, 42, 361–369. <https://doi.org/10.1016/j.bodyim.2022.07.008>
- Swami, V., Todd, J., & Tylka, T. L. (2022a). Measuring positive breast experience: Development and psychometric evaluation of the Breast Appreciation Scale (BrAS). *Body Image*, 43, 275–291. <https://doi.org/10.1016/j.bodyim.2022.09.007>
- Swami, V., Todd, J., Lazarescu, G., Bucur, V., & Vintilă, M. (2024). Translation and validation of a Romanian version of the Breast Appreciation Scale (BrAS). *Body Image*, 49, Article 101720. <https://doi.org/10.1016/j.bodyim.2024.101720>
- Swami, V., & Tovée, M. J. (2013). Men's oppressive beliefs predict their breast size preferences in women. *Archives of Sexual Behavior*, 42, 1199–1207. <https://doi.org/10.1007/s10508-013-0081-5>
- Swami, V., Tran, U. S., Barron, D., Afhami, R., Aimé, A., Almenara, C. A., Alp Dal, N., Amaral, A. C. S., Andrianto, S., Anjum, G., Argyrides, M., Atari, M., Aziz, M., Banai, B., Borowiec, J., Brewis, A., Cakir Kocak, Y., Campos, J. A. D. B., Carmona, C., & Voracek, M. (2020). The Breast Size Satisfaction Survey: Breast size dissatisfaction and its antecedents and outcomes in women from 40 countries. *Body Image*, 32, 199–217. <https://doi.org/10.1016/j.bodyim.2020.01.006>
- Swami, V., Tran, U. S., Stieger, S., Aavik, T., Abdollahpour Ranjbar, H., Adebayo, S. O., Afhami, R., Ahmed, O., Aimé, A., Akel, M., Al Halbusi, H., Alexias, G., Ali, K. F., Alp-Dal, N., Alsalthani, A. B., Álvares-Solas, S., Amaral, A. C. S., Andrianto, S., Apsden, T., & Voracek, M. (2023). Body appreciation around the world: Measurement invariance of the Body Appreciation Scale-2 (BAS-2) across 65 nations, 40 languages, gender identities, and age. *groups Body Image*, 46, 449–466. <https://doi.org/10.1016/j.bodyim.2023.07.010>
- Swami, V., Tran, U. S., Voracek, M., Aavik, T., Abdollahpour Ranjbar, H., Adebayo, S. O., Afhami, R., Ahmed, O., Aimé, A., Akel, M., Al Halbusi, H., Alexias, G., Ali, K. F., Alp-Dal, N., Alsalthani, A. B., Álvares-Solas, S., Amaral, A. C. S., Andrianto, S., Apsden, T., & Stieger, S. (2025). Self-compassion around the world: Measurement invariance of the Short Form of the Self-Compassion Scale (SCS-SF) across 65 nations, 40 languages, gender identities, and age groups. *Mindfulness*, 16, 1569–1596. <https://doi.org/10.1007/s12671-025-02560-5>
- Thornton, H., & Pillarisetti, R. (2008). 'Breast awareness' and 'breast self-examination' are not the same. What do these terms mean? Why are they confused? What can we do? *European Journal of Cancer*, 44(15), 2118–2121. <https://doi.org/10.1016/j.ejca.2008.08.015>
- Tylka, T. L., & Piran, N. (Eds.). (2019). *Handbook of positive body image and embodiment: Constructs, protective factors, and interventions*. Oxford University Press.
- Tylka, T. L., & Wood-Barcalow, N. L. (2015). The Body Appreciation Scale-2: Item refinement and psychometric evaluation. *Body Image*, 12, 53–67. <https://doi.org/10.1016/j.bodyim.2014.09.006>
- Vally, Z., D'Souza, C. G., Habeeb, H., & Bensumaidea, B. M. (2019). The factor structure and psychometric properties of an Arabic-translated version of the Body Appreciation Scale-2. *Perspectives in Psychiatry Care*, 55(3), 373–377. <https://doi.org/10.1111/ppc.12312>
- Worthington, R., & Whittaker, T. (2006). Scale development research: A content analysis and recommendations for best practice. *Counseling Psychologist*, 34, 806–838. <https://doi.org/10.1177/0011000006288127>
- Younes, M. (2019). *Beauty and the yield: Cosmetic surgery as a facet of neoliberal entrepreneurship in Lebanon*. Rosa-Luxemburg-Stiftung.
- Young, I. M. (1992). Breast experience: The look and the feeling. In D. Leder (Ed.), *The body in medical thought and practice* (pp. 215–230). Netherlands: Springer. [https://doi.org/10.1007/978-94-015-7924-7\\_13](https://doi.org/10.1007/978-94-015-7924-7_13)